Final Project Guidelines

Project presentations in-class on {6, 8, 11} May 2026
Project report due 12 May 2026 **(please submit a pdf)**

1 Overview

The final project will give you an opportunity to explore a new topic in numerical methods, going beyond the foundational topics we've covered in class. You can do this in two ways: (1) apply methods we've already covered to a *new simulation goal*, e.g., a more complex set of equations, which inevitably will lead to interesting computational observations and challenges; or (2) implement a more advanced numerical method. Of course, you can also do both (1) and (2). Examples of (1) include trying out a finite element method for a nonlinear reaction-diffusion equation, or a finite volume (or discontinuous Galerkin) method for the steady compressible Euler equations. Examples of (2) include studying and implementing higher-order finite element methods, or symplectic ODE integrators, or beyond. In general, you can build on material from either the first or second half of the semester. We will give many more examples and suggestions below.

2 Project proposal

First, do some brainstorming about topics, shaped by your own interests, our list of suggestions below, and any other sources of input that seem natural to you; certainly you can use Google Scholar or some LLM to search the current literature in computational science and numerical analysis for ideas. (Here, the *SIAM Journal on Scientific Computing* is a representative journal.)

Then, send a project proposal via email to both Andrey and Prof. Marzouk, no later than **Saturday 2 May 2026**. The project proposal need not be very formal or long; it can be an email with one paragraph describing what you'd like to do. In this email, please provide links to any paper(s) that are relevant to your idea. We can then iterate as needed to shape the goals and scope of the project into something both manageable and meaningful. Or the response might be to go right ahead.

3 Project report template

You should prepare a written project report of roughly 8 pages. LaTeX is preferred, and we suggest using the SIAM style files at `http://www.siam.org/journals/tex/siamltex.zip`. The report is due at 11:59 pm EDT on the last day of classes: **Tuesday, 12 May 2026**.

The report should provide a clear mathematical description of the problem being solved (e.g., what is the target ODE or PDE and what does it model); a clear mathematical presentation of the numerical methods you adopt; and a thorough analysis of your computational implementation and its performance. We suggest the following template:

- Abstract (one paragraph)

- Introduction: What is the focus of the project? A new (to you) numerical method, applied to some proof-of-concept problem? Or a more familiar method but extended to handle a more complex problem? What does the numerical method promise to achieve? What is interesting about the method and/or problem? Put everything in context, citing relevant literature.
- Mathematical models: In a self-contained way, describe the mathematical problem that you will solve numerically. Again, cite relevant literature as needed.
- Numerical method: Describe the numerical approach in a self-contained way and with full detail. As applicable, discuss convergence rates, computational cost, stability limits, scaling with grid resolution and other relevant problem input dimensions, etc. Are there any other properties of the numerical method than can be understood analytically?
- Computational experiments: Visualize your results and quantify errors in the numerical solution. As applicable, perform numerical studies of stability and accuracy; time evolution and error growth; convergence rates; computational cost (FLOPS, memory, wall time). Compare to competing algorithms.
- Conclusions: Summarize your key takeaways and directions for further refinement or exploration.
- Bibliography

4 Project presentations

We will schedule **oral presentations** of final projects in the last three class sessions. Signup for slots is on a first-come-first-served basis at this Google sheet. Each presentation should be 8 minutes in length, followed by 4 minutes for questions. Please focus your presentation on the essentials: the mathematical model, a discussion of the numerical methods and their properties, then some brief results. You'll find that eight minutes is not very long, so prepare only a handful slides.

These presentation sessions will give everyone in the class an opportunity to learn about the area you have explored, and hopefully broaden our collective perspective!

5 Topics

If you've come up with a project idea on your own, by all means pursue it! If you'd like some inspiration, however, here are some possibilities.

- Numerical integration in multiple dimensions: what can do you do beyond taking tensor products of one-dimensional quadrature rules? Look at cubature formulas, sparse grids.
- ODE integrators that conserve energy or other invariant quanties. Look at symplectic integrators of Hamiltonian systems.
- Dive deeply into Runge–Kutta methods, including implicit Runge–Kutta schemes.

- Integrating stochastic differential equations: analyze and implement schemes from Euler–Maruyama to higher order.

- Consider finite volume methods for a more realistic/advanced set of equations in conservation form, for instance the Euler equations for inviscid compressible flow.

- Related to the above: look at shock-capturing finite difference and finite volume schemes, and apply them to the 1-D Burgers equation.

- Implement a finite element method in two spatial dimensions. Implement a higher-order finite element method in two spatial dimensions.

- Look into mesh adaptivity and mesh refinement methods. Compare h-refinement and p-refinement approaches.

- Look into *a posteriori error estimation* schemes for finite element methods, and apply to a simple (e.g., 1-D diffusion) problem

- We will add more ideas here...